

SUMMARY

Backend and Infrastructure Engineer with experience building and operating production cloud services on AWS. Designed scalable PostgreSQL architectures, deployed microservices at scale, optimized CI/CD pipelines, and contributed to reliability-focused financial systems.

EDUCATION

• **PES University** 2021-25
Bachelor of Technology in Computer Science and Engineering

EXPERIENCE

- **Onecap Technologies - FinTech Startup** Jan 2025 – Present
Software Development Engineer Bangalore
- Built and operated 8 production microservices on AWS ECS Fargate, Lambda, ALB, S3, and Step Functions, for a financial reconciliation platform.
 - Designed and owned PostgreSQL architecture across platform services, including schema design, FDW-based cross-service data access, indexing strategies, and query optimization for high-volume reconciliation workloads.
 - Reduced Docker CI/CD build times by ~80% through dependency-layer caching and build optimization, accelerating deployment velocity across all backend services.
 - Automated long-running reconciliation workflows using AWS Step Functions, reducing manual operational effort through resilient retry, polling, and failure-handling mechanisms.
- **Center for Cloud Computing and Big Data** June – September 2024
Summer Research Intern Bangalore
- Built distributed data processing pipelines using PySpark and Hadoop for large-scale anomaly detection across edge network nodes, achieving 99% detection accuracy.
 - Designed automated remediation workflows integrating MQTT-based IoT infrastructure and CI/CD pipelines, significantly reducing operational response time.

PROJECTS

- **Lucid Care** Python, PyTorch
Multi-stage ML pipeline system for large-scale clinical data processing
- Built a multi-stage ML pipeline processing large-scale clinical datasets for prediction and recommendation workloads, achieving 96% model accuracy.
 - Integrated knowledge graphs and Graph RAG pipelines to improve contextual reasoning and reduce false positives in model outputs.
- **Sentinel Edge** PySpark, Hadoop, Python, PyTorch
Distributed anomaly detection and automated incident response at the network edge
- Developed distributed anomaly detection pipelines using PySpark, Hadoop, and PyTorch across edge computing environments, achieving 99% detection accuracy.
 - Automated incident response workflows for detected anomalies, eliminating manual remediation steps and improving system resilience.

SKILLS

Languages : Python, TypeScript, Java
Cloud : AWS (ECS Fargate, Lambda, ALB, S3, Step Functions, IAM, VPC, SES, CloudFront, ECR, API Gateway, Target Groups, Lightsail, Aurora and RDS)
Databases : PostgreSQL, MongoDB
Frameworks : FastAPI, Flask, Spring Boot
Containers & CI/CD : Docker, GitHub Actions
Big Data : PySpark, Hadoop

PUBLICATIONS

“ICCBDA 2025 — "LLM-Empowered Autonomous Edge AI for Anomaly Detection and Resolution in Distributed Systems"
“IntelliSys 2025 — "From Black Box to Bedside: Enhancing Interpretability and Trust in AI-Assisted Medical Systems"

ADDITIONAL DETAILS

Scrum Master, Onecap Technologies — Led sprint planning, backlog grooming, and delivery coordination across engineering teams.